find() in C++ STL

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C++ **find()** is a built-in function used to find the first occurrence of an element in the given range. It works with any container that supports iterators, such as arrays, vectors, lists, and more. In this article, we will learn about find() function in C++.

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<int> v = {1, 3, 6, 2, 9};

    // Search an element 6
    auto it = find(v.begin(), v.end(), 6);

    // Print index
    cout << distance(v.begin(), it);

    return 0;
}
```

Output

```
2
```

Syntax of find()

The std::find() is a C++ STL function defined inside **<algorithm>** header file.

```
find(first, last, val);
```

Parameters:

- **first**: Iterator to the first element of range.
- **last**: Iterator to the theoretical element just after the last element of range.
- **val**: Value to be searched.

Return Value:

- If the value is found in the range, returns an iterator to its position otherwise return end iterator.

Examples of find()

The following examples demonstrates the use of find() function for different cases:

Search for an Element in the Array

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int arr[5] = {1, 3, 6, 2, 9};

    // Search an element 6
    auto it = find(arr, arr + 5, 6);

    // Print index
    cout << distance(arr, it);
    return 0;
}
```

Output

```
2
```

In the above program, the **find()** function returns an iterator pointing to the value **6**, and we use the distance() function to print the position of the value 6, which is **2**.

Try to Find Element Which Not Present

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<int> v = {1, 3, 6, 2, 9};
```

```
    // Search an element 22
    auto it = find(v.begin(), v.end(), 22);

    // Check if element is preset
    if (it != v.end())

        // Print index
        cout << distance(v.begin(), it);
    else
        cout << "Not Present";

    return 0;
}
```

Output

```
Not Present
```

Explanation: The find() function returns the iterator to the end of the range if the element is not found. So, we compared returned iterator to the v.end() to check if the element is present in the vector.

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